

2.5 hours

Problem 4 (17JMO3)

We use signed areas. We claim that  $\frac{[DEF]}{[ABC]} = -2$ .

We let  $A = (1, 0, 0)$ ;  $B = (0, 1, 0)$ ;  $C = (0, 0, 1)$ . Let point  $P = (x: y: z)$ . We get that, since they are the intersections of cevians with sides,

$$D = (0: y: z);$$

$$E = (x: 0: z);$$

$$F = (x: y: 0)$$

We have,

$$\begin{aligned} \frac{[DEF]}{[ABC]} &= -2 \\ \Leftrightarrow \begin{vmatrix} 0 & \frac{y}{y+z} & \frac{z}{y+z} \\ \frac{x}{x+z} & 0 & \frac{z}{x+z} \\ \frac{x}{x+y} & \frac{y}{x+y} & 0 \end{vmatrix} &= -2 \\ \Leftrightarrow \frac{(2xyz)}{(x+y)(y+z)(z+x)} &= -2 \\ \Leftrightarrow -xyz &= (x+y)(y+z)(z+x) \\ \Leftrightarrow 0 &= (x+y)(y+z)(z+x) + xyz = (x+y+z)(xy+yz+zx) \end{aligned}$$

We show that  $xy + yz + zx = 0$ . Indeed,

However, we have,

$$(ABC): -a^2yz - b^2zx - c^2xy = 0$$

Since  $a = b = c \neq 0$ ,

$$\Leftrightarrow yz + zx + xy = 0$$

Therefore, we are done.

Problem 7 (Z07727CD)

We employ barycentric coordinates with  $A = (1, 0, 0)$ ;  $B = (0, 1, 0)$ ;  $C = (0, 0, 1)$ .

Since  $D$  is on the angle bisector, we write  $D = (a: t: c)$ . The line that contains  $D$  and the point at infinity  $(1: 0: -1)$  on  $AC$  is, clearly,

$$tx - (a + c)y + tz = 0$$

We intersect this with  $BC, AB$  to get,

$$R = (a + c : t : 0); S = (0 : t : a + c)$$

Since  $D$  is on the angle bisector, we write  $D = (a : t : c)$ . We now calculate  $P$ . Since  $DP$  is a cevian and  $P \neq C$ , we have, for some  $m$ ,

$$P = (a : t : m)$$

We now plug this into the formula for  $(ABC)$ .

$$-a^2tm - b^2am - c^2at = 0$$

This gives  $m = -\frac{c^2t}{at+b^2}$ . Therefore,

$$P = (a(b^2 + at) : t(b^2 + at) : -c^2t)$$

Let  $f$  be the homothety about  $B$  such that  $f(ABC) = RBS$ . Let,

$$f(P) = X; f(Q) = Y$$

Clearly,  $RQPS$  is cyclic if and only if  $AYXC$  is cyclic. We find that,

$$\frac{BC}{BS} = \frac{a + t + c}{a + c}$$

Therefore,

$$\begin{aligned} X &= \vec{B} + \frac{a + t + c}{a + c} (\vec{P} - \vec{B}) \\ &= \frac{a + t + c}{a + c} \vec{P} - \frac{t}{a + c} \vec{B} \end{aligned}$$

Factoring out the  $a + c$ ,

$$\begin{aligned} &= (a(b^2 + at)(a + t + c) : t(b^2 + at)(a + t + c) - t(a(b^2 + at) + t(b^2 + at) - c^2t) : -c^2t(a + t + c)) \\ &= (a(b^2 + at)(a + t + c) : ct(at + b^2 + ct) : -c^2t(a + t + c)) \end{aligned}$$

We now find the equation of  $(AXC)$ . Immediately, in the general form expression, the variable coefficients for  $x, z$  are zero, so, for some integer  $k$ ,

$$(AXC) : -a^2yz - b^2zx - c^2xy + (x + y + z)(ky) = 0$$

Plugging in  $X$ ,

$$\begin{aligned} k &= -\frac{a^2c^3t^2(at + b^2 + ct)(a + t + c) + ab^2c^2t(b^2 + at)(a + t + c)^2 - ac^3t(b^2 + at)(a + t + c)(at + b^2 + ct)}{ct(at + b^2 + ct)(a(b^2 + at)(a + t + c) + ct(at + b^2 + ct) - (a + t + c)(c^2t))} \\ &= \frac{ac(a + t + c)(act(at + b^2 + ct) + b^2(b^2 + at)(a + t + c) - c(b^2 + at)(at + b^2 + ct))}{(at + b^2 + ct)(a(b^2 + at)(a + t + c) + ct(at + b^2 + ct) - (a + t + c)(c^2t))} \\ &= \frac{ac(a + t + c)(-b^2c(at + b^2 + ct) + b^2(b^2 + at)(a + t + c))}{(at + b^2 + ct)(a(b^2 + at)(a + t + c) + ct(at + b^2 + ct) - (a + t + c)(c^2t))} \end{aligned}$$

$$\begin{aligned}
&= \frac{ab^2c(a+t+c)(-c(at+b^2+ct) + (b^2+at)(a+t+c))}{(at+b^2+ct)(a(b^2+at)(a+t+c) + ct(at+b^2+ct) - (a+t+c)(c^2t))} \\
&= \frac{ab^2c(a+t+c)(-act - b^2c - c^2t + ab^2 + b^2t + b^2c + a^2t + at^2 + act)}{(at+b^2+ct)(a(b^2+at)(a+t+c) + ct(at+b^2+ct) - (a+t+c)(c^2t))} \\
&= \frac{ab^2c(a+t+c)(-c^2t + ab^2 + b^2t + a^2t + at^2)}{(at+b^2+ct)(a(b^2+at)(a+t+c) + ct(at+b^2+ct) - (a+t+c)(c^2t))} \\
&= \frac{ab^2c(a+t+c)(-c^2t + ab^2 + b^2t + a^2t + at^2)}{(at+b^2+ct)(a+c)(a(b^2+at) + ct^2 - c^2t + b^2t + at^2 - ct^2)} \\
&= \frac{ab^2c(a+t+c)(-c^2t + ab^2 + b^2t + a^2t + at^2)}{(at+b^2+ct)(a+c)(ab^2 + a^2t - c^2t + b^2t + at^2)} \\
&= \frac{ab^2c(a+t+c)}{(at+b^2+ct)(a+c)}
\end{aligned}$$

This is symmetric in  $a, c$ , so, by symmetry, plugging in  $Y$  will give the same circle. Therefore, we are done.

#### Problem 8 (14IMO4)

Note that,  $\angle PAB = \angle BCA$  if and only if  $(ACP)$  is tangent to  $AB$  at  $A$ . Additionally,  $\angle CAQ = \angle ABC$  if and only if  $(ABQ)$  is tangent to  $AC$  at  $A$ .

Therefore, by Power of a Point,

$$\begin{aligned}
AC^2 &= QC \cdot BC; \\
AB^2 &= PB \cdot CB
\end{aligned}$$

We employ barycentric coordinates. Therefore,

$$\begin{aligned}
P &= (0 : a^2 - c^2 : c^2) = \left(0, \frac{a^2 - c^2}{a^2}, \frac{c^2}{a^2}\right) \\
Q &= (0 : b^2 : a^2 - b^2) = \left(0, \frac{b^2}{a^2}, \frac{a^2 - b^2}{a^2}\right)
\end{aligned}$$

We then reflect  $A$  over these to have,

$$M = \left(-1, \frac{2a^2 - 2c^2}{a^2}, \frac{2c^2}{a^2}\right); N = \left(-1, \frac{2b^2}{a^2}, \frac{2a^2 - 2b^2}{a^2}\right)$$

Therefore, cevian  $CN$  and  $BM$  intersect at  $(-a^2 : 2b^2 : 2c^2)$ . We can check that this lies on,

$$(ABC) : -a^2yz - b^2zx - c^2xy = 0$$

Plugging in,

$$-4a^2b^2c^2 + 2a^2b^2c^2 + 2a^2b^2c^2 = 0$$

As desired.

Problem 11 (23CZS3)

We employ barycentric coordinates with reference  $ABC$ .

We have,  $I = \left( \frac{a}{a+b+c}, \frac{b}{a+b+c}, \frac{c}{a+b+c} \right)$ , so,

$$E = \left( \frac{a-b-c}{a+b+c}, \frac{2b}{a+b+c}, \frac{2c}{a+b+c} \right) = (a-b-c:2b:2c)$$

Additionally, we know, since  $BD = s - b$ ;  $CD = s - c$ ,

$$D = (0:s-c:s-b) = (0:a+b-c:a-b+c)$$

Note that, clearly, the equation for  $DE$  is,

$$\begin{aligned} DE: 2(c-b)x + (a-b+c)y + (-a-b+c)z &= 0 \\ \Leftrightarrow (c-b)x + (s-b)y - (s-c)z &= 0 \end{aligned}$$

Note that  $(-1, 1, 1)$  is on the line.

We call this point  $A''$ . Note that  $A''$  is the reflection of  $A = (1, 0, 0)$  over the  $M = \left(0, \frac{1}{2}, \frac{1}{2}\right)$  midpoint of  $BC$ .

*Claim:*  $A'' \in (BHC)$

Note that reflecting over the midpoint of  $BC$  is the same as reflecting over  $BC$  then the perpendicular bisector of  $BC$ .

By reflecting the orthocenter,  $(BHC)$  is the reflection of  $(ABC)$  over  $BC$ , so  $A'$  the reflection of  $A$  of  $BC$  is on  $(BHC)$ . By symmetry,  $(BHC)$  is then the reflection of itself over the perpendicular bisector of  $BC$ . Therefore,  $A''$  the reflection of  $A'$  over this perpendicular bisector, equivalently the reflection of  $A$  over  $G$  is on  $(BHC)$  as desired.

*Claim:* If  $A''H$  is the diameter of  $(BHC)$ , we are done.

Note that,

$$\angle HFD = \angle HFA'' = 90^\circ$$

Therefore, if  $A''H$  is the diameter,  $F \in (BHC)$ .

We now employ complex coordinates to prove this.

Set  $(BHC)$  as the unit circle. Denote,  $a = H$ ;  $b = B$ ;  $c = C$ . Since  $A$  is the orthocenter of triangle  $ABC$ ,

$$A = a + b + c;$$

Therefore, since the midpoint of  $BC$  is  $\frac{b+c}{2}$ ,

$$A' = -a$$

Here, we are done.

19 clubs; 8 hours

Problem 12 (15SLG4)

We employ barycentric coordinates on  $ABC$ .

We let the second intersection of  $(BPMQ)$  with line  $AC$  apart from  $M$  be,

$$D = (m, 0, n)$$

Since,

$$CD \cdot CM = CQ \cdot CB$$

$$\Leftrightarrow bm \cdot \frac{b}{2} = CQ \cdot a$$

$$\Leftrightarrow \frac{b^2 m}{2a} = CQ$$

Therefore,

$$BQ = \frac{2a^2 - b^2 m}{2a}$$

Hence,

$$Q = (0 : b^2 m : 2a^2 - b^2 m);$$

Analogously,

$$P = (2c^2 - b^2 n : b^2 n : 0)$$

The line containing  $P = (2c^2 - b^2 n : b^2 n : 0)$  and the point at infinity on  $BC$ ,  $(0 : 1 : -1)$ , which is also the line  $PT$ , clearly has equation,

$$PT: b^2 n(x) + (b^2 n - 2c^2)(y) + (b^2 n - 2c^2)(z) = 0$$

Similarly,

$$QT: (b^2 m - 2a^2)(x) + (b^2 m - 2a^2)(y) + b^2 m(z) = 0$$

We find  $T$  that satisfies both equations. We have from the first equation,

$$2c^2(y + z) = b^2 n(x + y + z)$$

From the second,

$$2a^2(x + y) = b^2 m(x + y + z)$$

We let,

$$x + y = b^2 c^2 m;$$

$$x + y + z = 2a^2 c^2;$$

$$y + z = a^2 b^2 n$$

Therefore, we may have,

$$\begin{aligned} T &= (2a^2c^2 - a^2b^2n : a^2b^2n + b^2c^2m - 2a^2c^2 : 2a^2c^2 - b^2c^2m) \\ &= \left(1 - \frac{b^2n}{2c^2}, \frac{b^2n}{2c^2} + \frac{b^2m}{2a^2} - 1 : 1 - \frac{b^2m}{2a^2}\right) \end{aligned}$$

Since  $T \in (ABC)$ , we have,

$$-a^2yz - b^2xz - c^2xy = 0$$

The square of the distance from  $T = (x, y, z)$  to  $B = (0, 1, 0)$  is,

$$\begin{aligned} |TB|^2 &= -a^2(y-1)z - b^2xz - c^2x(y-1) \\ &= a^2z + c^2x \\ &= a^2\left(1 - \frac{b^2m}{2a^2}\right) + c^2\left(1 - \frac{b^2n}{2c^2}\right) \\ &= a^2 - \frac{b^2m}{2} + c^2 - \frac{b^2n}{2} \end{aligned}$$

Since  $m + n = 1$ ,

$$\Leftrightarrow |TB|^2 = a^2 + c^2 - \frac{b^2}{2}$$

On the other hand, the square of the distance from  $B = (0, 1, 0)$  to  $M = \left(\frac{1}{2}, 0, \frac{1}{2}\right)$  is,

$$\begin{aligned} |BM|^2 &= -a^2(-1)\left(\frac{1}{2}\right) - b^2\left(\frac{1}{2}\right)\left(\frac{1}{2}\right) - c^2\left(\frac{1}{2}\right)(-1) \\ &= \frac{1}{2}\left(a^2 + c^2 - \frac{b^2}{2}\right) \end{aligned}$$

Therefore,

$$\frac{|BT|}{|BM|} = \sqrt{\frac{|BT|^2}{|BM|^2}} = \sqrt{2}$$

### Problem 13 (13ESLG3)

Note that, if  $X$  exists,  $AX$  must be the  $A$ -symmedian. Therefore, if we let  $X'$  be the second intersection of the  $A$ -symmedian with  $(ARQ)$ , it is sufficient to prove  $X'$  is fixed.

We let  $P = (0, p, 1-p)$ . By similar triangles,  $R = (1-p, p, 0)$ ;  $Q = (p, 0, 1-p)$ . We find the equation of  $(ARQ)$ .

$$(ARQ): -a^2yz - b^2zx - c^2xy + (ux + vy + wz)(x + y + z) = 0$$

Plugging in  $A(1, 0, 0)$ , we have  $u = 0$ . Plugging in  $R$ ,

$$c^2p(p-1) + pv = 0 \Leftrightarrow v = c^2(1-p)$$

Plugging in  $Q$ ,

$$b^2p(p-1) + (1-p)w = 0 \Leftrightarrow w = b^2p$$

Therefore,

$$(ARQ): -a^2yz - b^2zx - c^2xy + (c^2(1-p)y + b^2pz)(x+y+z) = 0$$

Note that points on the  $A$ -symmedian have coordinates,  $(t: b^2: c^2)$ . We plug this in to acquire, at the intersection,

$$\begin{aligned} -a^2b^2c^2 - tb^2c^2 - tb^2c^2 + (c^2(1-p)b^2 + b^2pc^2)(t + b^2 + c^2) &= 0 \\ \Leftrightarrow -a^2b^2c^2 - tb^2c^2 - tb^2c^2 + (b^2c^2)(t + b^2 + c^2) &= 0 \end{aligned}$$

This is clearly not in terms of  $p$ , so we are done.

#### Problem 15 (12ELMO5)

We let  $D = (0, d, 1-d)$ ;  $E = (0, 1-d, d)$ . Points on the isogonal conjugate of  $AE$  contained inside  $ABC$  has coordinates  $(t: \frac{b^2}{1-d}: \frac{c^2}{d})$  for some positive  $t$ .

We may verify that the line from  $D = (0, d, 1-d)$  to  $(1: d-1: -d)$  the point at infinity on  $AE$  has equation,

$$(2d-1)x + (d-1)y + (d)z = 0$$

We plug in  $(t: \frac{b^2}{1-d}: \frac{c^2}{d})$  to get,

$$\begin{aligned} (2d-1)t - b^2 + c^2 &= 0 \\ \Leftrightarrow t &= \frac{b^2 - c^2}{2d-1} \end{aligned}$$

Therefore,

$$P = \left( \frac{b^2 - c^2}{2d-1}: \frac{b^2}{1-d}: \frac{c^2}{d} \right)$$

We now find  $P'$  the isogonal conjugate of this. We have,

$$P' = \left( \frac{a^2(2d-1)}{b^2 - c^2}: 1-d: d \right) = (a^2(2d-1): (1-d)(b^2 - c^2): d(b^2 - c^2))$$

We let the midpoint  $M$  of is  $(0, \frac{1}{2}, \frac{1}{2}) = (0: \frac{2a^2d - a^2 + b^2 - c^2}{2}: \frac{2a^2d - a^2 + b^2 - c^2}{2})$ . The sum of these coordinates is the same, so we may subtract to get,

$$\overrightarrow{MP'} = \left( -a^2(2d-1) : \frac{2a^2d - a^2 + b^2 - c^2}{2} - (1-d)(b^2 - c^2) : \frac{2a^2d - a^2 + b^2 - c^2}{2} - d(b^2 - c^2) \right)$$

On the other hand,

$$\overrightarrow{BC} = (0 : -1 : 1)$$

We can check that,

$$\overrightarrow{BC} \perp \overrightarrow{MP'}$$

$$\Leftrightarrow a^2 \left( - \left( \frac{2a^2d - a^2 + b^2 - c^2}{2} - d(b^2 - c^2) \right) + \left( \frac{2a^2d - a^2 + b^2 - c^2}{2} - (1-d)(b^2 - c^2) \right) \right) + b^2(-a^2(2d-1)) + c^2(a^2(2d-1)) = 0$$

$$\Leftrightarrow a^2(d(b^2 - c^2) + (d-1)(b^2 - c^2)) + b^2(-a^2(2d-1)) + c^2(a^2(2d-1)) = 0$$

We factor out  $2d-1$ ,

$$\Leftrightarrow a^2(b^2 - c^2) + b^2(-a^2) + c^2(a^2) = 0$$

This is clearly true. Therefore,  $P'$  lies on the perpendicular bisector of  $BC$ , so,

$$\angle P'BC = \angle P'CB$$

$$\Leftrightarrow \angle PBA = \angle PCA$$

As desired.

32 clubs; 11.5 hours